CSC 350 Assignment 1
Spring 2025

Due: Tuesday, January 28 (before class)

General Instructions

- Assignments are **due before the start of class** on the scheduled due date; generally, late assignments will not be accepted.
- Unless specified, **justify your answers always**. If justifications are missing or do not show reasonably how your answers are derived, 0 points will be assigned regardless of the correctness of the answers.
- Write **neatly** in case you write your answers on paper. **Especially, for table completion questions, if your answers are not given neatly in well-drawn tables, the answers won't be graded**.
- **This assignment must be done by yourself without any unauthorized assistance**. Write your full name and numeric university ID on the front page of your submission, along with the following pledge statement.

I pledge my honor that all parts of this assignment were done by me individually and without any unauthorized assistance or collaboration with others.

Q1. Precedence Rules

Write down the parenthesized version of each of the following expressions.

(a) $P \vee \neg Q \wedge R \rightarrow P \vee R \rightarrow \neg Q$

(b) $A \rightarrow B \vee \neg C \wedge D \wedge E \rightarrow F$

Q2. Well-Formed Formulas

For each of the expressions below, state whether it is a wff or not.

(a) $A \rightarrow B \vee \rightarrow \neg C \wedge D \wedge E \rightarrow F$

(b) $\neg \neg P \wedge Q$

(c) $(A \rightarrow B) \vee \neg (C \wedge D \wedge E) \rightarrow F \wedge (G \rightarrow \neg H)$

(d) $(A \rightarrow B \vee \neg C)(\wedge D \wedge E) \rightarrow F$

(e) $A \rightarrow (B \vee \neg (C \wedge D) \wedge E) \rightarrow F$

Q3. Syntax Tree

Draw the syntax tree for each of the following expressions.

(a) $(P \vee \neg Q) \wedge (\neg P \vee Q)$

(b) $P \vee \neg (Q \wedge R \rightarrow P) \vee (R \rightarrow \neg Q)$

Q4. Truth Tables

Use truth tables to verify the following equivalences.

(a) $(P \rightarrow Q) \equiv (\neg P \vee Q)$

(b) $\neg (P \wedge Q) \equiv (\neg P \vee \neg Q)$ (De Morgan's)

(c) $A \vee (A \wedge B) \equiv A$ (absorption)

(d) $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$ (distribution)

Q5. Quine's Method

Use Quine's method and basic equivalence to show whether each wff is a tautology, contradiction, or contingency.

(a) $A \wedge (A \rightarrow B) \rightarrow B$

(b) $(A \rightarrow B) \wedge (B \rightarrow C) \rightarrow (C \rightarrow A)$

Q6. Equivalences

Use equivalences to prove the following **(do not use truth tables)**.

$(P \rightarrow Q_1) \vee (P \rightarrow Q_2) \equiv P \rightarrow (Q_1 \vee Q_2)$

Hint: Start with the left-hand-side $((P \rightarrow Q_1) \vee (P \rightarrow Q_2))$. Apply a known equivalence to obtain an equivalent formula. Write down the equivalent formula and the equivalence used. Repeat this process until you obtain the right-hand side $(P \rightarrow (Q_1 \vee Q_2))$.